





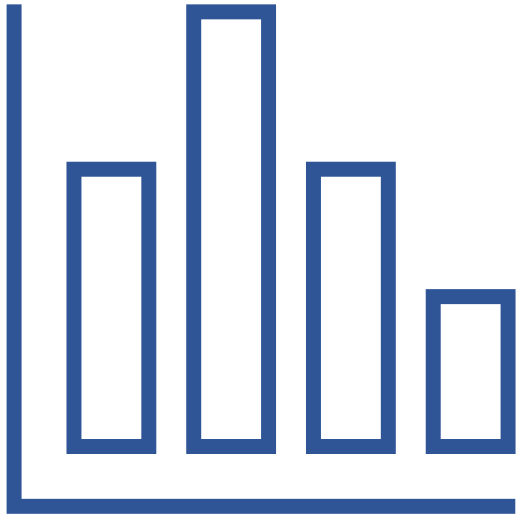
ENVRSUST 9014

Environmental Data Analysis and Visualization

Dr Paul Mensink









Collect & source data



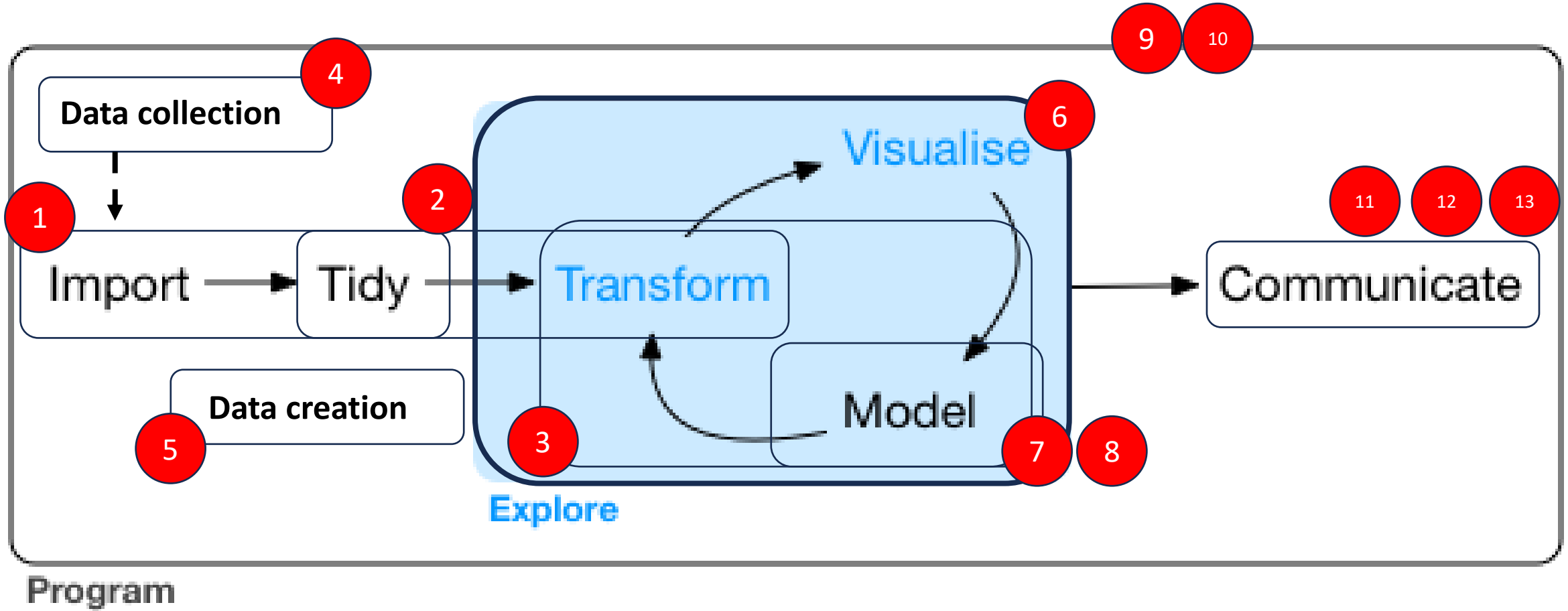
Organize & clean



Analyze, visualize & interpret

Course goals

Competent, confident and comfortable



R for Data Science
https://r4ds.had.co.nz/index.html

New skills are introduced each week that build on previous work.

1 INTRODUCE (BEFORE CLASS)



Watch short video lessons to introduce new skills and concepts.

2 PRACTICE (IN CLASS)



Work hands-on with peers and instructor to solve problems and build understanding.

3 TEST & REINFORCE (THROUGHOUT TERM)



- Mini-challenges
- Practice set questions

4 CONSOLIDATE & DEMONSTRATE (EXAMS)



Consolidate your learning and demonstrate understanding through a set of final exams.

5 APPLY (PROJECTS)

INDEPENDENT PROJECT



Source and analyze environmental data to answer your own question.

NPRI PROJECT

Addressing pressing national challenges by exploring the National Pollutant Release Inventory

Skills Confidence Impact



Data entry & organization



Data manipulation & visualization



Geospatial data analysis & mapping

Exploratory Data Analysis



Data entry & organization



Data manipulation & visualization



Geospatial data analysis & mapping



StoryMaps



BRIGHTSPACE OVERVIEW

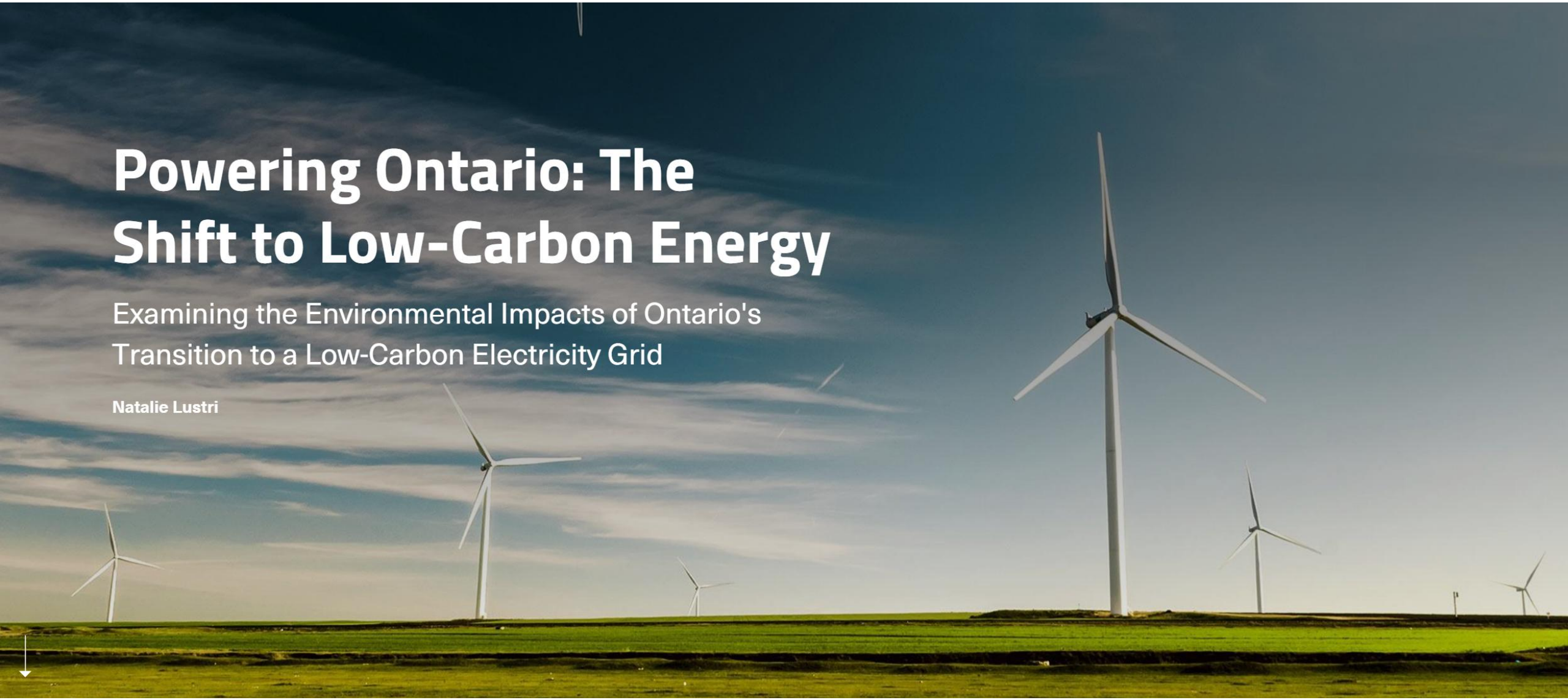


Agricultural Runoff and Water System Health

An Exploratory Examination of Agricultural Runoff and it's Impact to Waterbodies in Ontario

Bethany Reid-Smith

December 11, 2025



Powering Ontario: The Shift to Low-Carbon Energy

Examining the Environmental Impacts of Ontario's Transition to a Low-Carbon Electricity Grid

Natalie Lustri



Industrial Air Pollution near First Nations across Canada

An analysis of NPRI releases near communities from 2014 to 2023

By: Olivea Aziz

December 11, 2025